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CS-499

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4-2 Milestone Three: Enhancement Two: Algorithms and Data Structure

The artifact I selected for the Algorithms and Data Structures category is my *ProjectTwo.cpp* file from CS-300: Data Structures and Algorithms. I originally created this program during the CS-300 course as part of a project focused on building a Binary Search Tree (BST) to store, organize, and search university course information. The program loads course data from a file, inserts each course into the BST, prints the full course list in sorted order, and allows the user to look up individual courses and their prerequisites.

I chose this artifact for my ePortfolio because it demonstrates my understanding of core algorithmic concepts, including recursion, tree traversal, searching, and data organization. The BST structure is a classic example of how algorithms and data structures work together to solve real problems efficiently. Enhancing this artifact allowed me to showcase my ability to improve an existing algorithmic solution and add new functionality in a clean and maintainable way.

For my enhancement, I implemented a prefix-based search algorithm that allows users to search for all courses beginning with a specific prefix (such as “CS” or “MATH”). This required designing a recursive traversal that checks each node’s course number and collects matches into a vector. I also improved the in-order traversal so that it returns a sorted list of courses instead of printing directly to the console. This change makes the traversal more reusable and aligns with

good algorithmic design principles. These enhancements highlight my ability to work with recursive algorithms, manipulate tree structures, and improve the efficiency and flexibility of existing code.

In Module One, I planned to demonstrate my ability to design and evaluate computing solutions using algorithmic principles. With this enhancement, I met that goal. I implemented a new recursive search algorithm, improved the traversal logic, and strengthened the structure of the program. These changes directly support the outcome related to algorithms and data structures. At this time, I do not need to update my outcome-coverage plan, as the enhancement aligns well with what I intended to accomplish.

Enhancing this artifact helped me deepen my understanding of how algorithms operate within data structures and how small changes can significantly improve functionality. I learned how to design a recursive prefix-search algorithm that integrates cleanly with the existing BST structure. One challenge I faced was ensuring that the traversal logic remained efficient while also returning data in a reusable format. I also had to refactor parts of the original code to improve clarity and maintainability. Overall, this enhancement strengthened my confidence in working with recursive algorithms, tree-based data structures, and improving existing codebases in a thoughtful and structured way.